

Legal attribution - clear standards for sufficient digital evidence enable trustworthy attribution AND treaty verification

Alessandro Guarino 13, StudioAG – ICT Consulting & Engineering, “Digital Forensics in a Cyber Warfare Context,” StudioAG, March 2013, <http://www.studioag.pro/wp-content/uploads/2013/03/DigitalForensicsAndCyberwarfare.pdf>

1. Introduction Digital forensic has come a long way since its inception and has taken by now its place among the forensic sciences. Digital forensic techniques have now reached the state where they are less of an art and more like repeatable and well-documented scientific procedures. Work-flow phases have been codified in several guidelines developed by practitioners and also in developing international standards like ISO/IEC 27037, published probably 2012. This paper proposes a model under which digital forensics concepts developed along the years in a civilian context can be usefully applied to military operations, and particularly to cyber warfare. Scenarios in which digital forensics can be useful, and even necessary, include the attribution problem (of cyber attacks), treaty assurance, intelligence and counter-intelligence, both at tactical and strategic levels in the organization. At first sight the object of digital forensics in a warfare situation is very different from what is needed in the civilian world: here the “final product” is sound evidence, to be used in a court of law -civil or criminal- or at least that can be used, in principle. From there the need to assure the integrity of evidence from the very beginning and along all the chain of custody. In military contexts this necessity may seem unnecessary, compared to the need for actionable intelligence, often within strict time constraints, but contemporary warfare presents us with more and more legal concerns than the past. Contemporary international relations are more and more mad of supranational bodies with varying degrees of legal clout and status, from the United Nations to the International Court of Justice, various regional bodies and alliances, and a long history of international conventions currently in force. More and more we are faced with the phenomenon of nation-states presenting evidence of enemy conduct in order to justify a military action. If cyber warfare is to be a proper part of warfare, these evidence inevitably is and will be in digital form. Another example of digital forensics relevance is the problem of verification of possible future treaties regarding cyber warfare: this a thorny problem, and a very open one, currently debated by legal experts, military bodies and forensic analysts worldwide. It is quite easy to detect ICBM silos, not so easy to detect cyber weapons ready for action. International relations experts and country leaders are faced by the challenge of adapting concepts formulated long ago like aggression of a nation-state onto another, or combatant status, to the realities of cyber warfare. Also the relevance of non-national actors is very high in cyber warfare, by its very nature, complicating again the picture; we have only to think of the proxy problem, where non-national organizations or even individuals can conduct cyber activities on behalf of governments (knowingly or not). Legal concerns are mentioned because forensics is where the technical side meets the legal one: digital evidence is a tool in a legal process so it is to be conducted keeping in mind legal concerns like jurisdiction, lawful acquisition, and so on. The rest of this paper will present a brief overview of the field of digital forensics, as we see it now, followed by a short modelization of cyber warfare activities and context. After that a model matching digital forensics to the cyber warfare context will be proposed. 2. Digital Forensics What is digital forensics? We report here one of the most useful definitions of digital forensics formulated. It was developed during the first Digital Forensics Research Workshop (DFRWS) in 2001 and it is still very much relevant today: Digital Forensics is the use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations. (Pearson 2001) This formulation stresses first and foremost the scientific nature of digital forensics methods, in a point in time when it was transitioning from being a “craft” to an established field and rightful part of the forensic sciences. At that point digital forensics was also transitioning from being mainly practised in

separated environments such as law enforcement bodies and enterprise system administrators to a **unified field**.

Today this process is very advanced and it can be said that digital forensics principles, procedures and methods are shared by a large part of its practitioners, coming from different backgrounds (criminal prosecution, defence consultants, corporate investigators and compliance officers). Applying scientifically derived methods implies important concepts and principles to be respected when dealing with digital evidence. Among others we can cite: 1. Previous validation of tools and procedures. Tools and procedures should be validated by experiment prior to their application on actual evidence. 2. Reliability.

Processes should yield consistent results and tools should present consistent behaviour over time.

3. Repeatability. **Processes should generate the same results when applied to the same test environment**.

4. Documentation. **Forensic activities should be well-documented**, from the inception to the end of evidence life-cycle. On one hand **strict chain-of-custody** procedures **should be enforced** to assure evidence integrity and on the other hand complete documentation of every activity is necessary to ensure repeatability by other analysts.

5. Preservation of evidence – Digital evidence is easily altered and its integrity must be preserved at all times, from the very first stages of operations, to avoid spoliation and degradation. **Both technical** (e.g. hashing) **and**

organizational (e.g. clear accountability for operators) measures are to be taken.